

PROGRAM 7: Supplier Database

```
1 • create database supply;
2 • use supply;
3
4 • ⊖ create table supplier(
5     sid int primary key,
6     sname varchar(30),
7     city varchar(30));
8
9 • ⊖ create table parts(
10     pid int primary key,
11     pname varchar(30),
12     color varchar(30));
13
14 • ⊖ create table catalog(
15     sid int,
16     pid int,
17     cost int,
18     primary key (sid,pid),
19     foreign key (sid) references supplier (sid),
20     foreign key (pid) references parts (pid));
21
```

```

22 • insert into supplier values
23     (1,'acme widget','delhi'),
24     (2,'top traders','mumbai'),
25     (3,'best suppliers','chennai');
26
27 • insert into parts values
28     (101,'bolt','red'),
29     (102,'screw','white'),
30     (103,'nut','blue');
31
32 • insert into catalog values
33     (1,101,50),
34     (2,101,70),
35     (1,102,60),
36     (3,101,55),
37     (3,103,80);
38
39 # Find the pnames of parts for which there is some supplier.
40
41 • select distinct pname
42     from parts
43     where pid in (select pid from catalog);
..

```

Result Grid   Filter Rows: | Export:  | Wrap Cell Content: 

	pname
▶	bolt
	screw
	nut

45 # Find the snames of suppliers who supply every part.

46

47 • select sname

48 from supplier s

49 where not exists (select *

50 from parts p

51 where not exists (select *

52 from catalog c

53 where c.sid =s.sid and c.pid= p.pid)

54);

55

Result Grid



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Wrap Cell Content:

sname

55

56 # Find the snames of suppliers who supply every red part.

57

58 • select sname

59 from supplier s

60 where not exists (select *

61 from parts p

62 where p.color = 'red' and not exists (select * from catalog c

Result Grid



Filter Rows:

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Wrap Cell Content:

sname
acme widget
top traders
best suppliers

```

64
65 # Find the pname of parts supplied by Acme Widget Suppliers and by no one else.
66
67 • select pname
68   from parts p
69  where pid in (select pid
70                from catalog join supplier
71                where sname='acme widget') and pid not in (select pid
72                                                            from catalog join supplier
73                                                            where sname<>'acme widget');
74

```

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pname

```

75 # Find the sids of suppliers who charge more for some part than the average cost of that part
76
77 • select c.sid
78   from catalog c
79  where cost > ( select avg(cost)
80                from catalog
81                where pid =c.pid);
82

```

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sid
2

```

83 # For each part, find the sname of the supplier who charges the most for that part.
84
85 • select p.pname,s.sname
86   from catalog c
87  join supplier s on c.sid =s.sid
88  join parts p on c.pid=p.pid
89  where cost=(select max(cost)
90                from catalog
91                where pid =c.pid);
92
93

```

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pname	sname
screw	acme widget
bolt	top traders
nut	best suppliers